

BCA(I) — Math. Found. (101)**2025****(Session : 2024-27)****(Paper ID : 11016)**

Time : 3 hours

Full Marks : 80

Note: Candidates are required to give their answers in their own words as far as practicable. The questions are of equal value. Answer **any five** questions.

1. If $y = \frac{\tan^{-1}x}{1+x^2}$, prove that

$$y^{n+1} + \frac{2nx}{n+1} y^n + \frac{x^2(n-1)}{n+1} y^{n-1} = 0$$

2. If $u = \tan^{-1} \left(\frac{x^3 - y^3}{x - y} \right)$ then show that

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin(2u)$$

3. Determine the eigenvalues and eigenvectors of the matrix

$$A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

4. Find the inverse of the matrix using elementary row operation

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$$

5. Solve the following system of linear equation:

$$3x + 2y + z = 5$$

$$2x - y + 3z = 2$$

$$x + 4y - 2z = 16$$

6. Evaluate the following:

(a) $I = \int \frac{3x+2}{x^2+5x+1} dx$

(b) $I = \int_0^{\pi/2} \int_0^{1/\sqrt{2}} (x^2 + y^2) dy dx$

7. Solve: $(2x^2 + y^2 + 2xy + 3x + 3y) dx + (x^2 + y^2 + 4y) dy = e^{2x}$

8. Obtain expansion of $e^x \cos y$ in power of x and $\left(y - \frac{\pi}{2}\right)$ upto the second degree.

9. Find the maxima and minima of the function:

$$f(x, y) = x^3 + y^3 - 12x - 3y$$

10. Find the rank of the matrix:

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 & 1 \\ 2 & 4 & 1 & 3 & 0 \\ 3 & 6 & 2 & 5 & 1 \\ -4 & -8 & 1 & -3 & 1 \end{bmatrix}$$

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